25.9.2026

# Practical -8

## Nmap Live Host Discovery

**AIM:**

TO discover live hosts on a network using Nmap host-discovery scans - ARP, ICMP and TCP/UDP and understand protocol operations.

**INTRODUCTION:**

N-map (Network Mapper) is a free, open source network scanning tool created by Fyodor. It is used for mapping networks, discovering live hosts and identifying services running on them.

N-map can determine whether a host is alive using four different protocols.

- ARP - Link layer
- ICMP - Network Layer
- TCP - (Transport layer
- UDP - Transport Layer

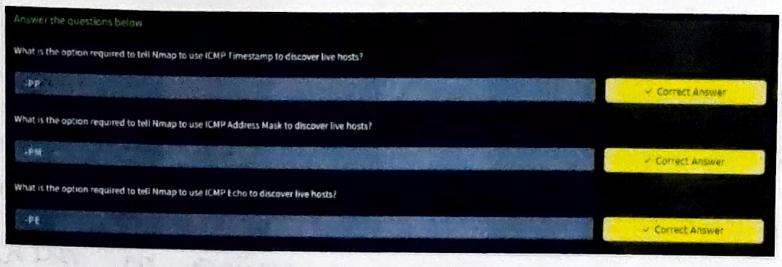
**Nmap host discovery using ARP**

Discovers hosts on same level subnet. It sends ARP request and live host respond with ARP replies. The -PR is used for ARP-based host discovery.



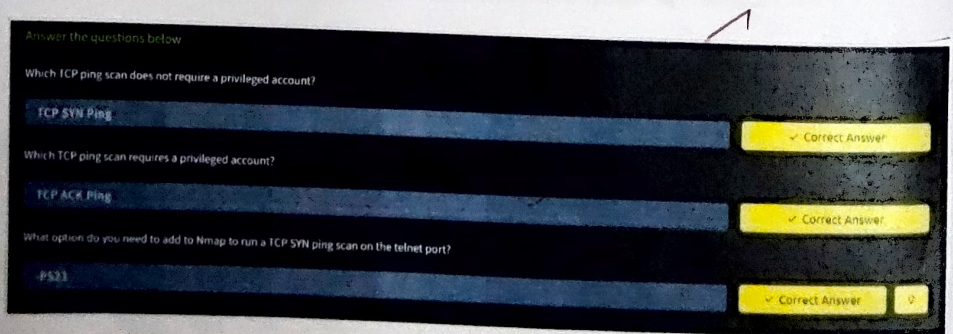
## Nmap host discovery using ICMP:

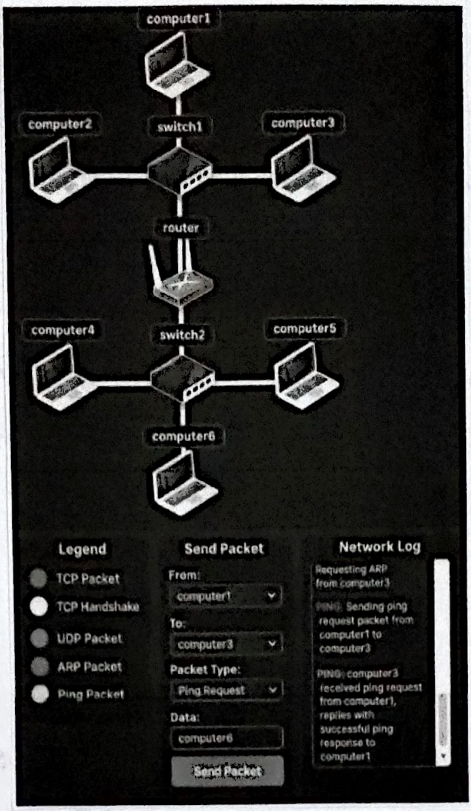
ICMP provides different req/reply mechanisms for discovering live host. It suggests ICMP echo (-PE), timestamp (-PP) and Address mask (-PM) scans.




## Nmap host discovery using TCP/UDP:

When ICMP is filtered, Nmap can use TCP and UDP packets to determine whether a host is online. TCP host includes (-PS) TCP SYN ping and TCP ACK ping (-PA) UDP uses -PU.






RESULT:

Thus live host in network were successfully discovered using N-map host discovery techniques such as ARP, ICMP and TCP, UDP ping scans. The different scanning methods are used to identify active hosts and verify their availability in network.